

Question	Answer	Mark	Mark scheme	Additional guidance
20 (a)	Shown	M1 M1 C1	for $\frac{n}{n+8}$ or starts to work with ratios, eg 3:7 forms equation and clears fractions, eg $10n = 7n + 56$ or $10n + 3(n+8) = 10(n+8)$ or equates $\frac{3}{10} = \frac{8}{x}$ or $\frac{3}{10} = \frac{8}{n+8}$ or continues to work with ratios, eg 3:7 = 24:56 gives the total sweets eg $\frac{80}{3}$ oe or number of red sweets $n = \frac{56}{3}$ oe or gives number of red as $\frac{56}{3}$ OR award 3 marks for a complete written argument, eg, $P(y) = \frac{3}{10}$ and there are 8 yellows. This cannot work as 3 is not a factor of 8 (and $\frac{3}{10}$ is in its simplest form)	Does not have to restate the $\frac{7}{10}$; giving a different probability will suffice

Paper: 1MA1/2H				
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20 (b)	28	P1 P1 P1 P1 A1	for $\frac{n}{n+8}$ and $\frac{n-1}{n+7}$ oe forms an appropriate equation, eg $\frac{n}{n+8} \times \frac{n-1}{n+7} = \frac{3}{5}$ for correctly forming a quadratic ready for solving, eg $an^2 + bn + c$ (= 0), $2n^2 - 50n - 168$ (= 0), $n^2 - 25n - 84$ (= 0) oe process to solve quadratic equation, ft a 3 term quadratic factorising eg $(n+3)(n-28)$ (=0) oe or completing the square or correct use of formula eg $\frac{- -25 \pm \sqrt{25^2 - 4 \times -84}}{2}$, $\frac{- -50 \pm \sqrt{50^2 - 4 \times 2 \times -168}}{2 \times 2}$ cao	Note we do not need to see “= 0”; just the LHS is sufficient. Award 0 marks for a correct answer with no supportive working.